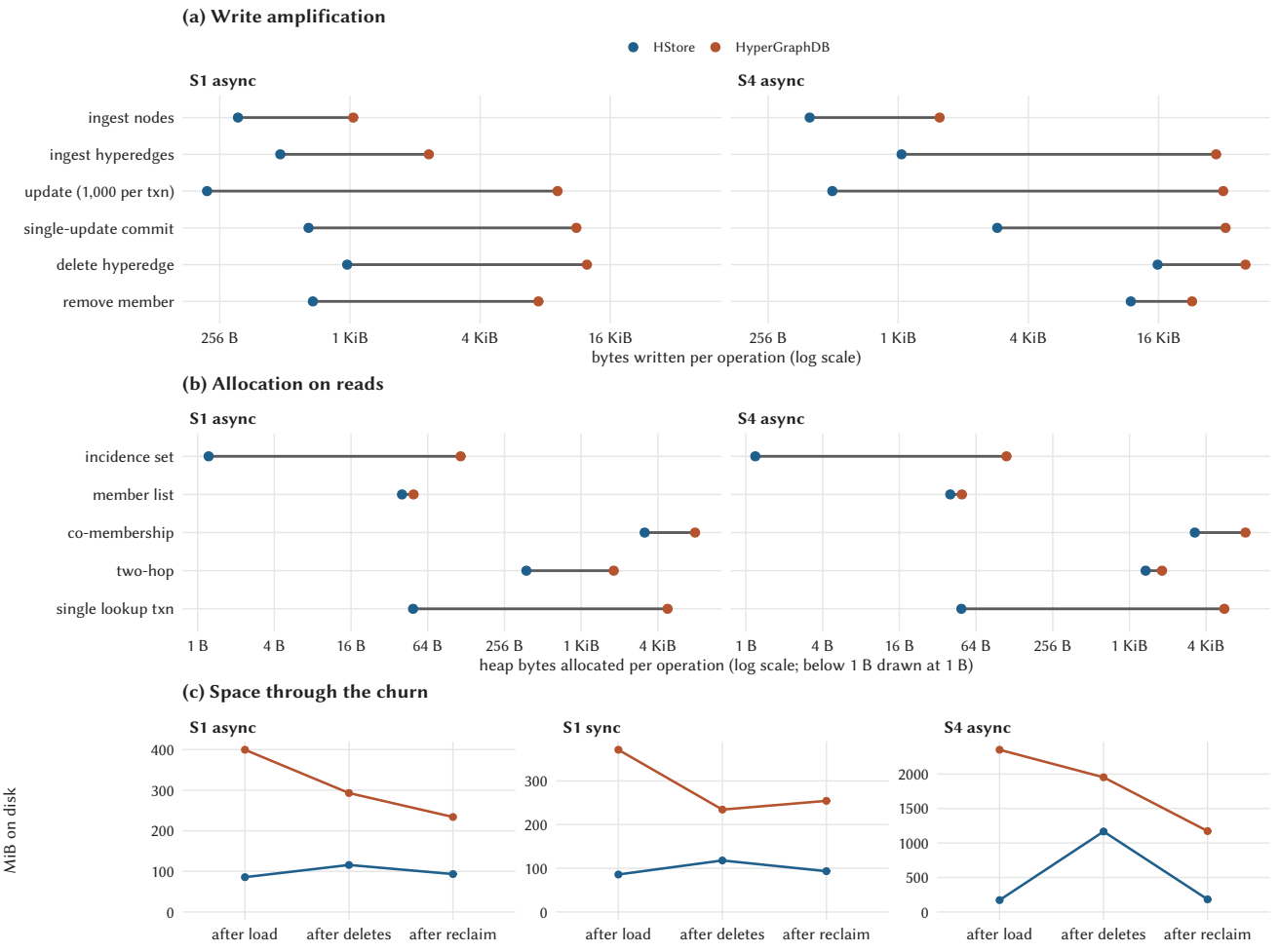
(a) Write amplification
HStore
HyperGraphDB
S1 async
S4 async
ingest nodes
ingest hyperedges
update (1,000 per txn)
single-update commit
delete hyperedge
remove member
256 B
1 KiB
4 KiB
16 KiB
bytes written per operation (log scale)
(b) Allocation on reads
S1 async
S4 async
incidence set
member list
co-membership
two-hop
single lookup txn
1 B
4 B
16 B
64 B
256 B
1 KiB
4 KiB
heap bytes allocated per operation (log scale; below 1 B drawn at 1 B)
(c) Space through the churn
S1 async
S1 sync
S4 async
MiB on disk
0
100
200
300
400
500
1000
1500
2000
after load
after deletes
after reclaim